Database backups older than seven days did not exist

INC-5608 · found during DR-2026-Q2 · S. Caduff (Infrastructure Resilience) · closed 2026-05-22

Jira issue	INC-5608 · actions RES-410 to RES-413
Severity	SEV-2 (major) — latent data-loss exposure; no customer impact
Found	2026-05-12 14:20 CEST, DR exercise step 7 (restore from 30 days ago)
Resolved	2026-05-15 17:00 CEST
Financial impact	CHF 0 — no data was lost and no restore was needed in production
Affected systems	backup-orchestrator, bucket hr-prod-db-backups

Description

The quarterly DR exercise on 12 May 2026 includes restoring the customer database as it stood 30 days earlier. The restore failed at step 7 because no backup older than seven days existed in the backup bucket. Our backup policy requires 35 days of daily backups. The nightly backup job had succeeded every night; the backups it wrote were being deleted a week later.

For 82 days, from 19 February to 12 May, any corruption or deletion discovered more than a week after the fact could not have been recovered from backup. No such event occurred in that period.

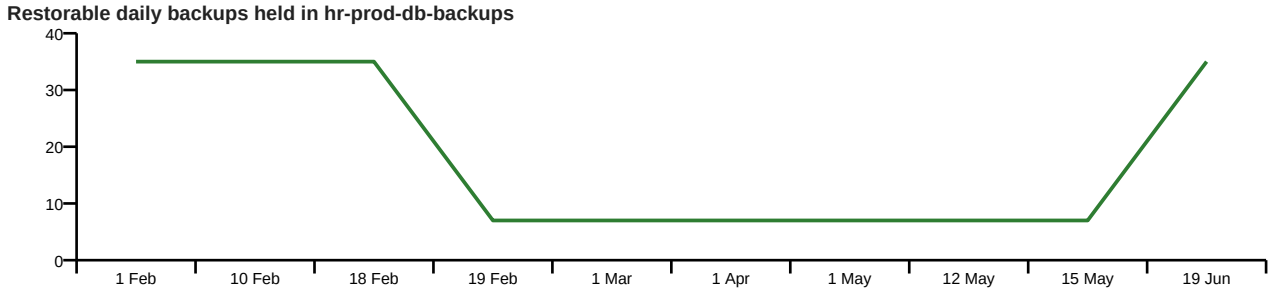

Figure 1 — retention falls from 35 to 7 on 19 February and returns to 35 once the lifecycle rule is removed and a month of backups has accumulated again.

Root cause

On 19 February 2026 a storage cost-reduction change, FINOPS-118, applied a standard lifecycle rule to every bucket whose name begins `hr-prod-`. The rule expires objects seven days after creation. It was written for log buckets, and the backup bucket `hr-prod-db-backups` matched the prefix. The change was reviewed by FinOps, who did not know the bucket held backups; the bucket carried no tag saying so.

The exposure lasted 82 days because backup monitoring checks only that last night's backup was written. Nothing verified that the oldest required backup still existed.

Timeline

When	What happened
2026-02-19 10:30	FINOPS-118 applied: 7-day expiry on all hr-prod-* buckets (31 buckets).
2026-02-26	First backup deleted by the rule.
2026-05-12 14:20	DR step 7 fails: oldest backup is dated 5 May.
2026-05-12 14:45	Lifecycle rule found on the bucket. INC-5608 raised.
2026-05-12 15:30	Rule removed from hr-prod-db-backups and from two other buckets holding backups.
2026-05-13	Second, independent backup copy to the DR region started, to cover the 35-day window while it refills.
2026-05-15 17:00	Controls in place; exposure reduced to the refill period. Resolved.
2026-05-22	Review held; incident closed.

Action plan

Jira	Action	Owner	Due	Status
RES-410	Tag every bucket holding backups data-class=backup and deny lifecycle rules on that tag by policy.	S. Caduff	2026-05-29	Done
RES-411	Daily check that the oldest required backup (day 35) exists and is restorable; page on failure.	A. Novak	2026-06-05	Done
RES-412	Lifecycle changes by prefix require a listing of every matched bucket in the change record.	FinOps (J. Meier)	2026-06-12	Done
RES-413	Move the 30-day restore from the quarterly DR exercise into a weekly automated test.	S. Caduff	2026-06-26	Done